

PROTOCOLS - EFI LOADED IMAGE

This section defines `EFI_LOADED_IMAGE_PROTOCOL` and the `EFI_LOADED_IMAGE_DEVICE_PATH_PROTOCOL`. Respectively, these protocols describe an Image that has been loaded into memory and specifies the device path used when a PE/COFF image was loaded through the EFI Boot Service `LoadImage()`. These descriptions include the source from which the image was loaded, the current location of the image in memory, the type of memory allocated for the image, and the parameters passed to the image when it was invoked.

9.1 EFI Loaded Image Protocol

9.1.1 EFI_LOADED_IMAGE_PROTOCOL

Summary

Can be used on any image handle to obtain information about the loaded image.

GUID

```
#define EFI_LOADED_IMAGE_PROTOCOL_GUID \
    {0x5B1B31A1, 0x9562, 0x11d2, \
     {0x8E, 0x3F, 0x00, 0xA0, 0xC9, 0x69, 0x72, 0x3B}}
```

Revision Number

```
#define EFI_LOADED_IMAGE_PROTOCOL_REVISION 0x1000
```

Protocol Interface Structure

```
typedef struct {
    UINT32                Revision;
    EFI_HANDLE             ParentHandle;
    EFI_System_Table       *SystemTable;

    // Source location of the image
    EFI_HANDLE             DeviceHandle;
    EFI_DEVICE_PATH_PROTOCOL *FilePath;
    VOID                   *Reserved;

    // Image's load options
    UINT32                LoadOptionsSize;
    VOID                   *LoadOptions;
```

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```
// Location where image was loaded
VOID                               *ImageBase;
UINT64                             ImageSize;
EFI_MEMORY_TYPE                    ImageCodeType;
EFI_MEMORY_TYPE                    ImageDataType;
EFI_IMAGE_UNLOAD                   Unload;
} EFI_LOADED_IMAGE_PROTOCOL;
```

Parameters

Revision

Defines the revision of the EFI_LOADED_IMAGE_PROTOCOL structure. All future revisions will be backward compatible to the current revision.

ParentHandle

Parent image's image handle. NULL if the image is loaded directly from the firmware's boot manager. Type EFI_HANDLE is defined in *Services — Boot Services*.

SystemTable

The image's EFI system table pointer. Type EFI_SYSTEM_TABLE defined in *EFI System Table*.

DeviceHandle

The device handle that the EFI Image was loaded from. Type EFI_HANDLE is defined in *Services — Boot Services*.

FilePath

A pointer to the file path portion specific to DeviceHandle that the EFI Image was loaded from. EFI_DEVICE_PATH_PROTOCOL is defined in *EFI Device Path Protocol*.

Reserved

Reserved. DO NOT USE.

LoadOptionsSize

The size in bytes of LoadOptions.

LoadOptions

A pointer to the image's binary load options. See the OptionalData parameter in the *Load Options* section of the Boot Manager chapter for information on the source of the LoadOptions data.

ImageBase

The base address at which the image was loaded.

ImageSize

The size in bytes of the loaded image.

ImageCodeType

The memory type that the code sections were loaded as. Type EFI_MEMORY_TYPE is defined in *Services — Boot Services*.

ImageDataType

The memory type that the data sections were loaded as. Type *EFI_MEMORY_TYPE* is defined in *Services — Boot Services*.

Unload

Function that unloads the image - see [Section 9.1.2](#).

Description

Each loaded image has an image handle that supports EFI_LOADED_IMAGE_PROTOCOL. When an image is started, it is passed the image handle for itself. The image can use the handle to obtain its relevant image data stored in the

EFI_LOADED_IMAGE_PROTOCOL structure, such as its load options.

9.1.2 EFI_LOADED_IMAGE_PROTOCOL.Unload()

Summary

Unloads an image from memory.

Prototype

```
typedef
EFI_STATUS
(EFIAPI *EFI_IMAGE_UNLOAD) (
    IN EFI_HANDLE          ImageHandle,
    );
```

Parameters

ImageHandle

The handle to the image to unload. Type `EFI_HANDLE` *Driver Model Boot Services*

Description

The Unload() function is a callback that a driver registers to do cleanup when the UnloadImage boot service function is called.

If the Unload() function pointer in an EFI_LOADED_IMAGE_PROTOCOL instance is NULL, the image does not support unload.

Status Codes Returned

EFI_SUCCESS	The image was unloaded.
EFI_INVALID_PARAMETER	The <i>ImageHandle</i> was not valid.

9.2 EFI Loaded Image Device Path Protocol

9.2.1 EFI_LOADED_IMAGE_DEVICE_PATH_PROTOCOL

Summary

When installed, the Loaded Image Device Path Protocol specifies the device path that was used when a PE/COFF image was loaded through the EFI Boot Service LoadImage().

GUID

```
#define EFI_LOADED_IMAGE_DEVICE_PATH_PROTOCOL_GUID \
    {0xbc62157e, 0x3e33, 0x4fec, \
     {0x99, 0x20, 0x2d, 0x3b, 0x36, 0xd7, 0x50, 0xdf}}
```

Description

The Loaded Image Device Path Protocol uses the same protocol interface structure as the *EFI Device Path Protocol* defined in Chapter 10. The only difference between the Device Path Protocol and the Loaded Image Device Path Protocol is the protocol GUID value.

The Loaded Image Device Path Protocol must be installed onto the image handle of a PE/COFF image loaded through the EFI Boot Service `LoadImage()`. A copy of the device path specified by the *DevicePath* parameter to the EFI Boot Service `LoadImage()` is made before it is installed onto the image handle. It is legal to call `LoadImage()` for a buffer in memory with a NULL *DevicePath* parameter. In this case, the Loaded Image Device Path Protocol is installed with a NULL interface pointer.